

Machine Learning I: Hands-On 8 - Worked Solutions

8.1 Ensembles

The true targets and the four classifier predictions are:

	ex1	ex2	ex3	ex4	ex5	ex6	ex7	ex8	ex9	ex10
Target	1	0	0	0	0	1	0	1	0	0
A	1	1	0	0	0	1	1	1	0	0
B	1	0	1	0	0	1	0	0	1	0
C	1	0	0	1	1	1	0	1	0	1
D	0	1	1	0	0	0	0	1	0	0

1. Individual classifier accuracy

For a binary classifier,

$$\text{Accuracy} = \frac{\text{number of correct predictions}}{\text{number of examples}}.$$

Comparing each row with the true targets gives:

Classifier	Correct predictions	Accuracy
A	8/10	0.80
B	7/10	0.70
C	7/10	0.70
D	6/10	0.60

Classifier A is the best **individual classifier by accuracy** in this example.

2. Majority-vote ensembles

For each example, the ensemble outputs the class predicted by at least two of its three members.

Ensemble	Majority-vote predictions for ex1–ex10	Correct	Accuracy
A+B+C	1, 0, 0, 0, 0, 1, 0, 1, 0, 0	10/10	1.00
A+B+D	1, 1, 1, 0, 0, 1, 0, 1, 0, 0	8/10	0.80
A+C+D	1, 1, 0, 0, 0, 1, 0, 1, 0, 0	9/10	0.90
B+C+D	1, 0, 1, 0, 0, 1, 0, 1, 0, 0	9/10	0.90

The best ensemble is A+B+C, with 100% accuracy.

3. Why can the ensemble outperform its members?

The best individual classifier, A, reaches 80% accuracy, whereas A+B+C reaches 100%. The improvement is possible because the three classifiers do not make the same mistakes on the same examples. When one member is wrong, the other two can still form a correct majority.

ML insight: diversity matters

A majority-vote ensemble benefits from **useful disagreement**: its members should be individually competent but should not systematically make the same errors. Majority voting cannot repair an error shared by most members.

8.2 Imbalanced Domain Learning

There are 20 examples: 18 negatives and 2 positives. Class 1 is the positive/minority class.

The confusion-matrix terms are:

Term	Meaning
TP	predicted 1 and the true class is 1
FP	predicted 1 but the true class is 0
FN	predicted 0 but the true class is 1
TN	predicted 0 and the true class is 0

		Predicted	
		0	1
Actual	0	TN	FP
	1	FN	TP

1. Confusion matrices

Read each prediction row against the true targets.

Classifier A

A predicts class 0 for every example. Therefore:

- $TP = 0$: no positive example is detected;
- $FP = 0$: it never predicts a false positive;
- $FN = 2$: both positive examples are missed;
- $TN = 18$: all negative examples are correctly classified.

Classifier B

B predicts both positives correctly, but also predicts four negatives as positive:

- $TP = 2$;
- $FP = 4$;
- $FN = 0$;
- $TN = 14$.

Classifier C

C detects one of the two positives and creates no false positives:

- $TP = 1$;
- $FP = 0$;
- $FN = 1$;
- $TN = 18$.

Classifier D

D detects one positive and incorrectly flags two negatives:

- $TP = 1$;
- $FP = 2$;
- $FN = 1$;
- $TN = 16$.

The four confusion matrices can therefore be summarized as:

Classifier	TN	FP	FN	TP
A	18	0	2	0
B	14	4	0	2
C	18	0	1	1
D	16	2	1	1

2. Accuracy, precision, recall, and F1-score

Use

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN},$$

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN},$$

$$F_1 = 2 \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}.$$

Classifier A

$$\text{Accuracy} = \frac{18}{20} = 0.90,$$

but it never predicts the positive class, so

$$\text{Precision} = 0, \quad \text{Recall} = 0, \quad F_1 = 0.$$

Classifier B

$$\text{Accuracy} = \frac{14 + 2}{20} = 0.80,$$

$$\text{Precision} = \frac{2}{2 + 4} = \frac{1}{3} \approx 0.33,$$

$$\text{Recall} = \frac{2}{2 + 0} = 1.00,$$

$$F_1 = 2 \frac{(1/3)(1)}{1/3 + 1} = 0.50.$$

Classifier C

$$\text{Accuracy} = \frac{18 + 1}{20} = 0.95,$$

$$\text{Precision} = \frac{1}{1 + 0} = 1.00, \quad \text{Recall} = \frac{1}{1 + 1} = 0.50,$$

$$F_1 = 2 \frac{(1)(0.5)}{1 + 0.5} \approx 0.67.$$

Classifier D

$$\text{Accuracy} = \frac{16 + 1}{20} = 0.85,$$

$$\text{Precision} = \frac{1}{1 + 2} = \frac{1}{3} \approx 0.33, \quad \text{Recall} = \frac{1}{1 + 1} = 0.50,$$

$$F_1 = 2 \frac{(1/3)(0.5)}{1/3 + 0.5} = 0.40.$$

The final comparison is:

Classifier	Accuracy	Precision	Recall	F_1
A	0.90	0.00	0.00	0.00
B	0.80	0.33	1.00	0.50
C	0.95	1.00	0.50	0.67
D	0.85	0.33	0.50	0.40

3. Which classifier should be selected?

There is no context-free answer.

If **false negatives are extremely costly**, Classifier B is the natural choice among these four because its recall is 1.00: it finds both positive examples. The cost is four false positives and lower overall accuracy.

If **false positives are very costly**, but detecting at least some positive examples is required, Classifier C is attractive. It produces no false positives and detects one of the two positives. The trade-off is recall 0.50: one positive example is missed.

Classifier A is the key counterexample to selecting by accuracy alone. It achieves 90% accuracy despite detecting **none** of the minority examples. The high accuracy is produced simply by exploiting the majority-class frequency.

ML insight: metrics encode different error preferences

Accuracy measures overall correctness. Recall focuses on missed positives, precision focuses on false alarms, and F_1 balances precision and recall symmetrically. None of these metrics defines a universally optimal classifier without first specifying which mistakes matter.